

Dulitha Prasanna Kulathunga

Postdoctoral Research Associate

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📍 Chapel Hill, NC

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Summary

- Computational and theoretical chemist and biophysicist working at the interface of molecular simulation, machine learning, and scientific discovery. I combine molecular dynamics, quantum chemistry, and enhanced-sampling simulations with active machine learning and data-driven optimization to find promising molecules, reaction conditions, and protein sequences with as few costly experiments or simulations as possible.

Education

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| Iowa State University & Ames National Laboratory , Physical Chemistry | Ames, IA |
| • Thesis advisors: Prof. Theresa L. Windus and Prof. Davit Potoyan. | 2021 – Apr 2026 |
| • Dissertation: <i>Exploration without Exhaustion</i> , defended April 10, 2026. | |
| University of Colombo , Computational Chemistry | Colombo, Sri Lanka |
| • Completed 2020. Advisor: Prof. Samantha Weerasinghe. | |
| • CUFSAA Gold Medal for Best Performance in Computational Chemistry (2019). | |
| University of Colombo School of Computing (UCSC) , Information Technology | Colombo, Sri Lanka |
| • Completed 2019. | |

Experience

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| University of North Carolina at Chapel Hill , Postdoctoral Research Associate, Miao Lab | Department of Pharmacology, Chapel Hill, NC |
| • Advisor: Prof. Yinglong Miao. | 2026 – present |
| • Gaussian accelerated molecular dynamics (GaMD) enhanced-sampling simulations of RNA and RNA–ligand systems. | 1 year |
| • Developing AI-driven models for RNA-targeted drug discovery and virtual screening. | |
| Iowa State University & Ames National Laboratory , Graduate Research Assistant | Ames, IA |
| • AMLRO : built an open-source active learning framework for single- and multi-objective reaction optimization, with a Flask/Streamlit GUI for open-loop use and SHAP-based explainability. Applied it to Paal-Knorr flow synthesis, closed-loop diglycolamide (DGA) flow synthesis, and aldehyde-selective Wacker oxidation in a 7-dimensional parameter space. | 2021 – Apr 2026 |
| • CyRxnOpt : co-developed a unified interface for benchmarking reaction optimization algorithms, integrated into the RxnRover automation ecosystem. | 5 years |
| • ALSEBO : developed a coevolution-informed Bayesian optimization framework over a generative latent sequence landscape for data-efficient protein design; designed novel green fluorescent protein variants validated by MD simulations. | |
| • Enzyme thermal adaptation : studied serine proteases and cold-adapted β -glucosidases from psychrophilic to thermophilic organisms using molecular dynamics, bioinformatics, and unsupervised machine learning. | |
| • Collaborated with Oak Ridge National Laboratory and the Critical Materials Innovation Hub (CMI) on multi-objective optimization of molten salt electrochemical graphitization. | |
| Iowa State University , Graduate Teaching Assistant | Ames, IA |
| • General chemistry laboratories and recitations (Chem 163/163L, 167L, 177L). | 2021 – 2022 |
| | 1 year |

University of Colombo, Teaching Assistant

- Computational chemistry laboratory courses for undergraduate and M.Sc. students.

Colombo, Sri Lanka

2020 – 2021

1 year

University of Colombo, Undergraduate Research Assistant

- Compared all-atom (AMBER99SB-ILDN) and united-atom (KBFF) force fields for interactions between β -amyloid and model amino acids with unsaturated side chains, using GROMACS.

Colombo, Sri Lanka

2019 – 2020

1 year

Open Source Projects

AMLRO (Active Machine Learning Reaction Optimizer)

Active learning framework for single- and multi-objective optimization of chemical reaction conditions.

- Closed-loop (automated) and open-loop (human-in-the-loop) optimization, with a web GUI.
- Explainable AI with SHAP to rank the influence of reaction variables.
- Released as DOE software: [doi:10.11578/dc.20260205.1](https://doi.org/10.11578/dc.20260205.1).

AMLRO GUI

Web interface for running AMLRO active learning optimization campaigns without writing Python (Flask API backend, React/TypeScript frontend).

ALSEBO (Active Learning Sequence Exploration via Bayesian Optimization)

Generative protein design by Bayesian optimization in a learned latent space.

- Variational autoencoder (VAE) latent space for protein sequences.
- Coevolution-informed Bayesian optimization to propose new variants with few measurements.
- Demonstrated on avGFP fluorescence ([bioRxiv 2026](#)).

CyRxnOpt

Common Python interface for reaction optimization algorithms, usable from a CLI or inside self-optimizing platforms.

Awards

Ames National Laboratory Innovation Incentive Award (for CyRxnOpt)

2026

Ames National Laboratory

Klaus Ruedenberg Theoretical Chemistry Award

2025

Department of Chemistry, Iowa State University

Drtina Chemistry Graduate Student Travel Award

2025

Received in 2024 and 2025.

Department of Chemistry, Iowa State University

Edward V. Sayre Chemistry Scholarship

2024

Department of Chemistry, Iowa State University

CUFSAA Gold Medal for Best Performance in Computational Chemistry

2019

University of Colombo

Skills

Programming & Apps**Machine Learning**

Molecular Simulation

Computational Chemistry & Cheminformatics